

Minesweeper Sep 30 Testing Document

EECS 581 Spring 2026

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Date: 30 Sep, 2026

Overview

Software Name: Implemented Minesweeper

Tester: Warren Tan

Test Date: 30 Sep, 2026

Test Cases

Test ID	Description	Method	Expected Result	Actual Result	Status
TC-01	0 Mines	Set the initial mines to 0	reject input	reject input	Pass
TC-02	10 Mines	Set the initial mines to 10	10 mines are placed	10 mines are placed	Pass
TC-03	10.4 Mines	Set the initial mines to 10.4	reject input	11 mines are placed. Game is unwinnable.	Fail
TC-04	10.5 Mines	Set the initial mines to 10.5	reject input	11 mines are placed. Game is unwinnable.	Fail
TC-05	20 Mines	Set the initial mines to 20	20 mines are placed	20 mines are placed	Pass
TC-06	99 Mines	Set the initial mines to 99	reject input	reject input	Pass
TC-07	First-Click Protection	Set the initial mines to 20. Start the game. Click a cell.	first cell is not a mine	first cell is not a mine	Pass
TC-08	First-Click Protection w/ Flag	Set the initial mines to 20. Start the game. Flag a cell. Click a cell.	first cell is not a mine	first cell is not a mine	Pass

Test ID	Description	Method	Expected Result	Actual Result	Status
TC-09	Board Size1	Run buildGrid(10, 10, bombs)	10x10 grid created	10x10 grid created	Pass
TC-10	Board Size2	Run buildGrid(10, 11, bombs)	10x11 grid created	User is directed to game screen. No grid is created	Fail
TC-11	Board Size3	Run buildGrid(11, 10, bombs)	11x10 grid created	User is directed to game screen. No grid is created	Fail
TC-12	Board Size4	Run buildGrid(11, 11, bombs)	11x11 grid created	11x11 grid created	Pass
TC-13	Board Size5	Run buildGrid(9, 9, bombs)	9x9 grid created	9x9 grid created	Pass
TC-14	User Wins	Finish the game with 10 bombs	User is directed to win screen	User is directed to win screen	Pass
TC-15	Lose Music	Lose a game, then restart to title screen.	Game music is paused on loss, then active on title screen	Game music is paused on loss, then active on title screen	Pass

Test ID	Description	Method	Expected Result	Actual Result	Status
TC-16	Easy AI Solver	Set the initial mines to 15. Choose "vs AI". Set AI Difficulty to "Easy". Toggle "AI plays automatically". Start game.	AI chooses tiles at random	AI chooses tiles at random	Pass
TC-17	Medium AI Solver	Set the initial mines to 15. Choose "vs AI". Set AI Difficulty to "Medium". Toggle "AI plays automatically". Start game.	AI with smart flagging and opening (detailed in design documents)	AI solved with basic flagging and opening logic	Pass
TC-18	Hard AI Solver	Set the initial mines to 15. Choose "vs AI". Set AI Difficulty to "Hard". Toggle "AI plays automatically". Start game.	Medium AI with 1-2-1 pattern solving.	Solved the board while utilising the 1-2-1 pattern.	Pass

Test ID	Description	Method	Expected Result	Actual Result	Status
TC-19	Local Multiplayer Bombs	Set the initial mines to 15. Choose "2 player". Start game.	Game is started with 15 mines placed.	15 mines are placed.	Pass
TC-20	Local Multiplayer Flags	Set the initial mines to 15. Choose "2 player". Player 1 flags a tile. Player 2 attempts to remove Player 1's flag.	Flag is placed and cannot be altered by opponent.	Flag is placed. Player 2 cannot remove Player 1's flag.	Pass
TC-21	Local Multiplayer Instant Death	Set the initial mines to 15. Choose "2 player". Select "Instant Death". Select tiles until mine is selected.	Game is ended. Opposing player wins.	Game is ended. Player 2 selected the mine and lost.	Pass
TC-22	Local Multiplayer Point Tracking	Set the initial mines to 15. Choose "2 player". Select "Points". Play game as normal while tracking point changes.	-3 points for mine hit. +1 for flagging a real mine. -2 for flagging a safe tile.	-3 points for mine hit. No points for flagging mine or safe tiles.	Partial Fail
TC-23	Local Multiplayer Point Win	Set the initial mines to 15. Choose "2 player". Select "Points". Play game as normal.	The player with the greater points wins.	Player with the greater point count wins game.	Pass